

Canonical Spin–Geometry and the Primitive Symplectic Atlas

Volume 0. The volume's strongest exact result: the four primitives are canonical charts on one radial-spinor phase plane — and the audit that keeps the imported machinery honest.

About this book. Rebuilt as standalone R Theory from the former T6 essay-book, the "canonical" book of the Projection Craft series — its declared job was to make the primitive functions canonical coordinates on the radial-spinor phase space. That core is proved. Around it sits heavy imported machinery (Dirac–Coulomb theory, the first-order Einstein–Hilbert action, Maxwell, rotation, Einstein–Cartan, Kerr), and this book labels every layer: what is proved internally, what is conditional on import, and what is not established here. The "promotion" of earlier results is explained precisely: it means *restriction to a specialized setting and reuse of existing proofs* — never a status upgrade and never new evidence.

Status key. **EXACT** proved mathematics · **CONDITIONAL** exact after a declared import or contract · **IMPORT** accepted physics taken as input, not derived · **OBSTRUCTION** a proved or retained negative result.

Depends on: [Book 0](#); [Book 20](#) (§4's two-angle structure); the internal radial Dirac–Coulomb paper.

§1 Spin transport

§2 Prüfer pair

§3 Symplectic atlas

§4 Defect action

§5 Gauge & rotation

§6 Kerr

§7 Spin obstruction

§8 Audit

§1 — Radial coframe and spin transport (imported machinery, labeled)

The book's canonical apparatus starts from accepted background: **IMPORT** the radial coframe e^a (flat at $\eta=0$, with defect $\delta_e = e^0 - dr$), the torsion-free spin connection, and the Dirac spinor Ψ in the projected chart. Two declared contracts carry the whole book: **C6a**, the radial two-component chart $\Psi \rightarrow (r, F, G, \dots)$, and **C6b**, that F and G are independent on open sets. Two obstructions are kept permanently: the

spherical-to-Cartesian map is not a global diffeomorphism (origin and polar axis excluded), and C6a/C6b fix the *reading* of the geometry but *not* the physical meaning of F and G — importing the radial Dirac–Coulomb paper promotes the chart, never the book's own assumptions.

§2 — The Prüfer canonical pair (exact conditional)

The transport operator on (F, G) has exact zero trace, which yields the exact radial invariant

$$F F' + G G' = -A^2/(2r)$$

and a canonical pair: the Prüfer angle $\theta = \arg(G + iF)$ and the spin density $A^2 = F^2 + G^2$. Under C6a/C6b the pair is regular on open sets and the spin phase space carries

$$\omega_{\text{spin}} = 4 \, dF \wedge dG$$

— exactly canonical. This is exact *after* importing the radial Dirac dynamics; the symplectic form is a conditional theorem, not a derivation of Dirac theory.

Canonical spin phase space CONDITIONAL

Under C6a/C6b and the imported radial Dirac equations, (Θ, A^2) is a canonical pair on the radial-spinor phase plane with symplectic form $4 \, dF \wedge dG$. OBSTRUCTION Does not derive the Dirac equation, spin- $\frac{1}{2}$, the spin-statistics connection, or the fine-structure constant.

§3 — The primitive symplectic atlas (the volume's strongest exact result)

The centerpiece. The four primitive functions are **canonical charts on the same (F, G) plane** — four Darboux-compatible coordinate systems with a common spin one-form:

$$p_{\text{sxp}} = 2F^2, \quad p_{\text{srx}} = -2G^2, \quad p_{\text{cxp}} = (F-G)^2, \quad p_{\text{crx}} = -(F+G)^2$$

$$p_Q dQ = -2G dF + 2F dG$$

and the common symplectic form $\omega_{\text{spin}} = 4 dF \wedge dG$, already established in §2. The atlas maps cover all four quadrants; the reciprocal and complementary identities are exact differential identities. On the circular Dirac–Coulomb branch the atlas charts meet the radial paper's spinor-ratio theorem, so the atlas variables and the spinor-ratio results are cross-checked against the same imported dynamics. No new physics is claimed: the primitives were already canonical coordinate expressions; the atlas shows they are canonical *together*.

Primitive symplectic atlas **EXACT**

The four primitive momentum charts with common one-form and common symplectic form constitute an exact internal theorem, conditional only on the declared C6a/C6b chart and the imported radial dynamics. The chart transition identities, reciprocal relations, and amplitude-ratio readings are exact. This is a representation result: it changes no prediction of the imported theory.

§4 — The first-order spherical defect action (conditional reconstruction)

Now the imported gravitational machinery. **IMPORT** the first-order Einstein–Hilbert action in the spherical defect variables $(\delta_e, \delta_f, \delta_\omega)$, with a regularized radial measure $\eta^2(r)dr$ on (r_{UV}, ∞) and the standard asymptotically flat falloff conditions declared. Exact reductions under these imports: the gravitational defect kinetic structure; the boundary term; and the vanishing of the on-shell defect action in the massless flat defect-free limit. What remains outside the reduction: the regularization contract, the falloff conditions, and the accepted Einstein–Hilbert machinery itself — conditional reconstruction, not a derivation of GR. The mass parameter m appearing in the defect action is imported through the contract; the defect action does not compute it.

CONDITIONAL **Spherical defect reduction.** Exact reduction of the imported first-order Einstein–Hilbert action in the defect variables under declared regularization and falloff. **OBSTRUCTION** Does not derive the Einstein field equations, Newton’s constant, or the mass parameter.

§5 — Gauge, rotation, and spin: conditional channels

Three more channels, each exact after its own import:

- **Gauge (Maxwell as a channel):** after the gauge import, the radial electric charge appears as the cyclic radial momentum p_Φ conjugate to the radial gauge phase — the Maxwell field-strength dynamics and the coupling e are imported; the book supplies the canonical reading, not the theory.
- **Rotation:** angular momentum enters as the canonical momentum p_Ω conjugate to frame dragging — exact in the imported Kerr-geometry channel.
- **Spin (Einstein–Cartan):** the axial contorsion channel is exactly auxiliary: the torsion equation is algebraic and the torsion is solved out in favor of the spin current — so the spin sector carries no independent propagating degree beyond the imported one.

Each channel is a representation of imported physics in the book's canonical coordinates. None derives its import.

§6 — Kerr/Kerr–Newman half-angle identities (with guards)

On the imported Kerr geometry, the horizon radii satisfy exact half-angle identities in the book's coordinates — and the same identities extend to Kerr–Newman after the electromagnetic import. The guards are part of the result: the identities live on the declared equatorial/chart domain (coordinate singularities excluded), they are consequences of the imported metric, and they establish no new black-hole physics. The spherical-spin obstruction (§7) is the reason these identities do not become a spin law.

CONDITIONAL **Kerr/Kerr–Newman identities.** Exact coordinate identities on the imported geometries, valid on the declared regular domain. **OBSTRUCTION** No new black-hole physics; no spin dynamics derived.

§7 — The spherical-spin obstruction

The book's hardest negative, and its most load-bearing. The spherical symmetry used throughout forces the total spin current to integrate to zero over the sphere — a spherically symmetric configuration cannot carry net spin. The canonical spin apparatus of §§1–3 therefore **cannot be promoted to a theory of spin in the spherical sector**. This is not a defect of the atlas; it is a theorem about the domain: the spin variables are exact internal coordinates of the radial-spinor phase plane, and the spherical obstruction says exactly where they stop being physics. Any future spin program must begin by leaving spherical symmetry.

OBSTRUCTION **Spherical spin cancellation.** Spherical symmetry kills net spin current. The atlas is exact where it applies and silent where it cannot speak.

§8 — The does-not-establish ledger and book closure

The source book's §6.XV is preserved verbatim in spirit: the does-not-establish ledger. This book does not derive: the Dirac equation; spin- $\frac{1}{2}$; the fine-structure constant; the Coulomb interaction; the Einstein field equations; Newton's constant; the Maxwell equations; the charge e ; the Kerr metric; the Einstein–Cartan spin coupling; particle masses; couplings; or new observables. "Promotion" of earlier results (Books 20–21, the radial paper) means restriction to a specialized setting with reuse of existing proofs — a status-preserving operation, never a status upgrade and never new evidence. The Yang–Mills attribution from the source preamble is **not** repeated: the cited result is absent from the cited text (see the [Volume 0 provenance note](#)).

Book 22 promotion audit

Exact internal theorems:	spin invariant, Prüfer pair, primitive symplectic atlas, atlas transition identities.
Exact after declared import:	Kerr/KN identities, defect-action reduction, gauge/rotation/spin channels.
New equation of motion:	none.
New coupling or mass parameter:	none.
New invariant observable:	none.

Prediction difference: $\Delta_{\text{op}} = \emptyset$. C0 retained.

Book 22 closure EXACT

The four primitives are canonical charts on one radial-spinor phase plane — the primitive symplectic atlas is the strongest exact result of Volume 0, and it changes nothing it did not already change: it is a representation theorem inside imported physics. With the obstructions kept (spherical spin cancellation, coordinate-domain guards, the does-not-establish ledger), $\Delta_{\text{op}} = \emptyset$ and C0 — *Projection Craft changes representation but not physics* — is retained, undefeated, and still falsifiable.

Forward boundary. The book's own reopening rule: C0 is defeated only by a future fully specified Class-C model with a stated measurement contract — a new invariant observable that survives the redundancies this volume catalogued. The spherical-spin obstruction marks where any spin program must start: outside spherical symmetry.

[← Series index](#) · [Volume 0 overview](#) · [← Fermion Composition](#)

A Volume 0 book of the R Theory rewrite. Source: the former T6 essay-book, rebuilt standalone. Status discipline: exact / conditional / import / obstruction — nothing promoted beyond its support. $\Delta_{\text{op}} = \emptyset$ throughout.